

OCEAN WARMING

Big Data and Cloud Computing for Climate

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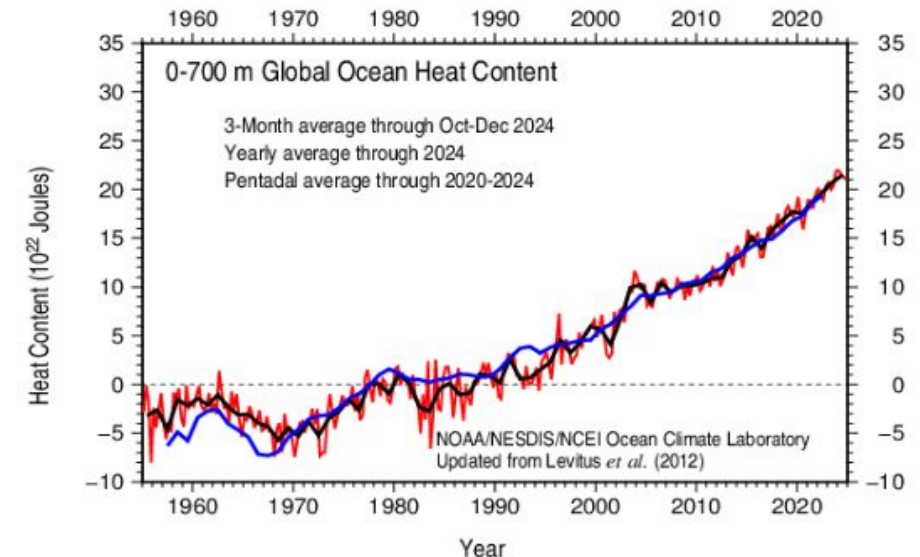
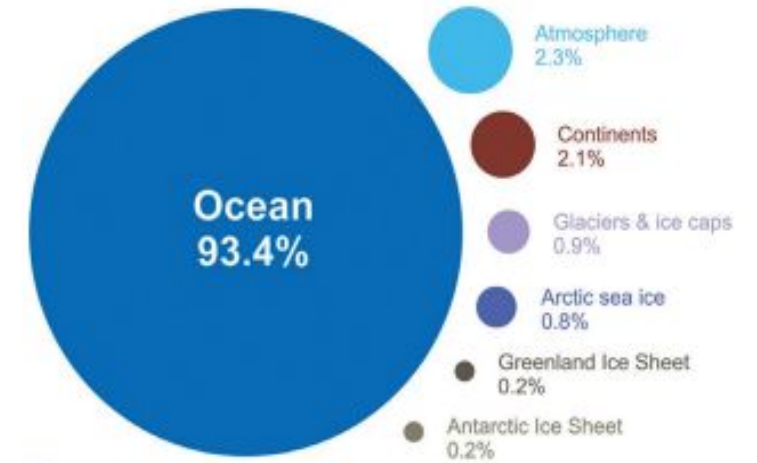
Louis Cabanis

Background: Ocean Warming

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- **Why has ocean warming become a threat?**
 - The ocean plays a role in climate regulation
 - Observational data has shown a significant increase in ocean heat content
- **Scientific basis and knowledge:**
 - Global oceans have absorbed more than 90% of the excess heat trapped by human-induced greenhouse gases
 - Ocean warming is a direct consequence of global warming
- **Trends in ocean heat content (OHC)**
 - Analysis of global OHC over time reveals a clear trend of warming
 - Ongoing rise in ocean temperatures has far-reaching implications

Where is global warming going?



Background: Ocean Heat Content (OHC)

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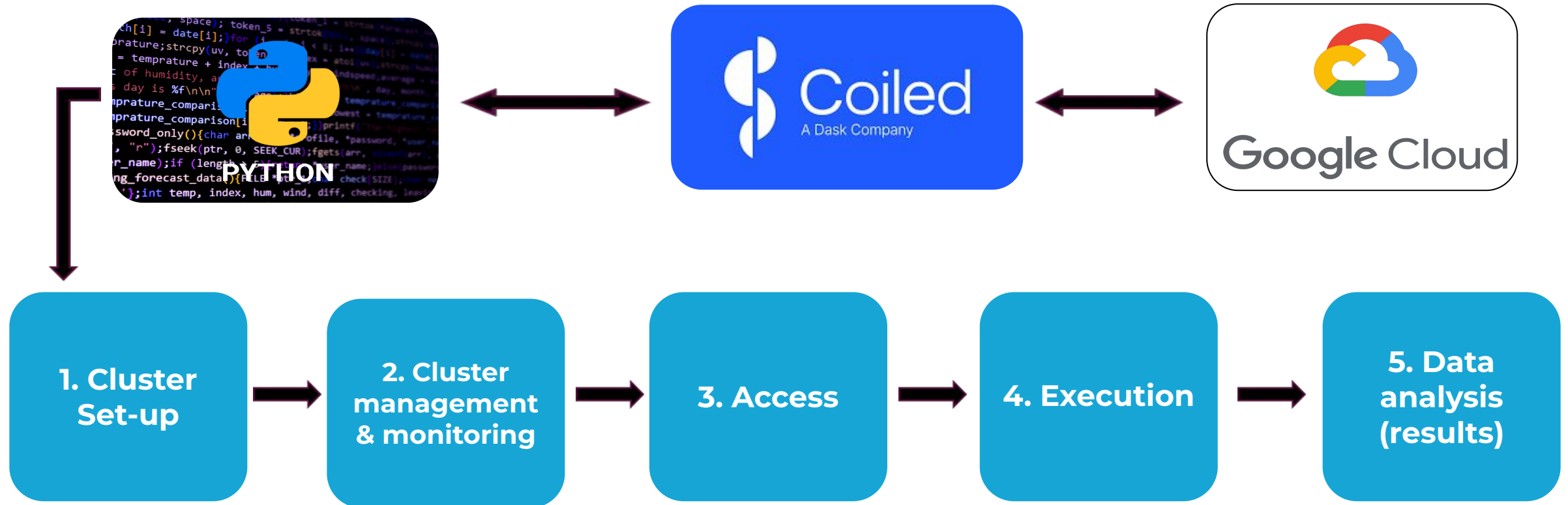
- What is Ocean Heat Content (OHC)?

$$OHC(x, y, z, t) = \rho \cdot C_p \cdot T(x, y, z, t) \cdot \delta V(x, y, z)$$

- Where:
 - ρ = Density of seawater(1035 kg/m³)
 - C_p = Specific heat capacity of seawater(4 186 J/(kg · K))
 - $T(x,y,z,t)$ = Temperature of seawater at location x,y,z and time t
 - $\delta V(x,y,z)$ = Volume element at a given location and depth
- The OHC is important for:
 - Climate regulation
 - Sea-level rise
 - Marine ecosystem impact
 - Weather systems

Data Analysis: Methodology

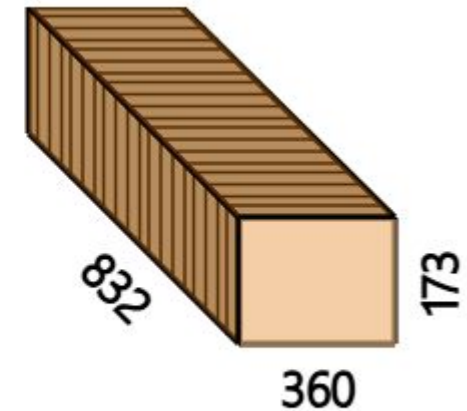
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Data Analysis: Dataset

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- EN4 Dataset is a global interpolation of all available ocean in-situ observations providing a comprehensive picture of the ocean's thermal and salinity state
 - **Size:** 52 Gb
 - **Spatial resolution:** 1-degree horizontal resolution (173 latitude points × 360 longitude points), covering the globe.
 - **Temporal resolution:** Monthly data - from 1950 to 2019
 - **Vertical layers:** 42 depth levels (up to 5350m)

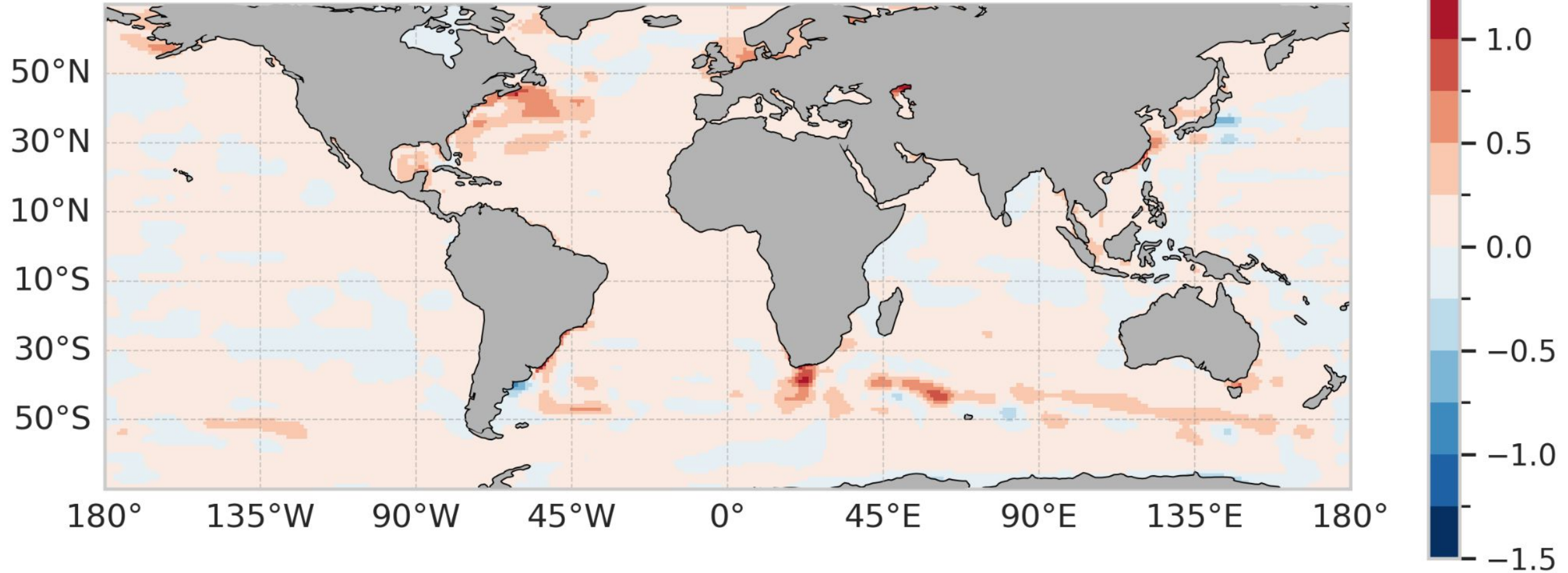


An array representing the temperature at a certain depth

Results: Ocean Heat Content (OHC)

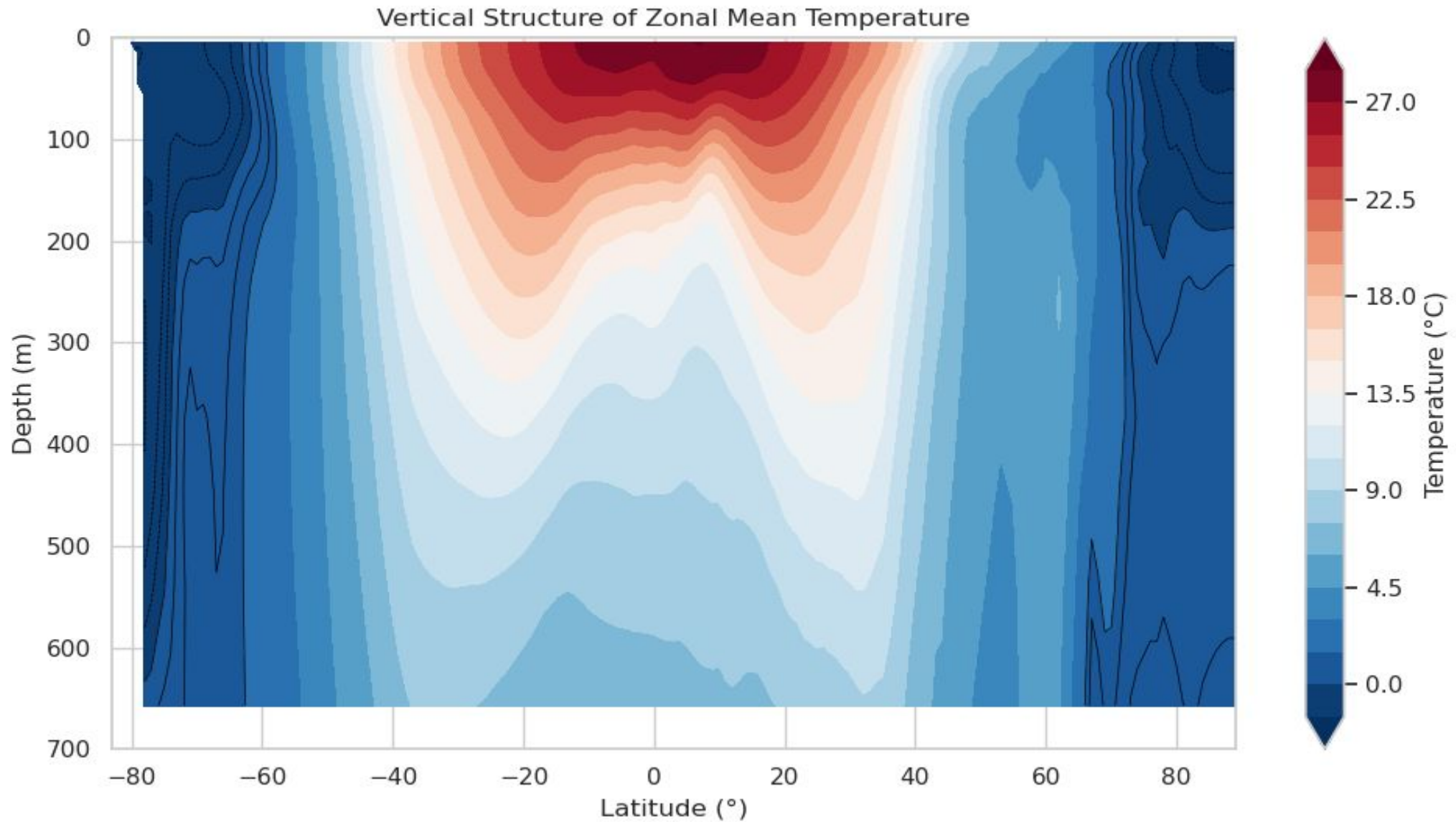
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Local 0-700m temperature trend in mK/decade (1950 to 2019)

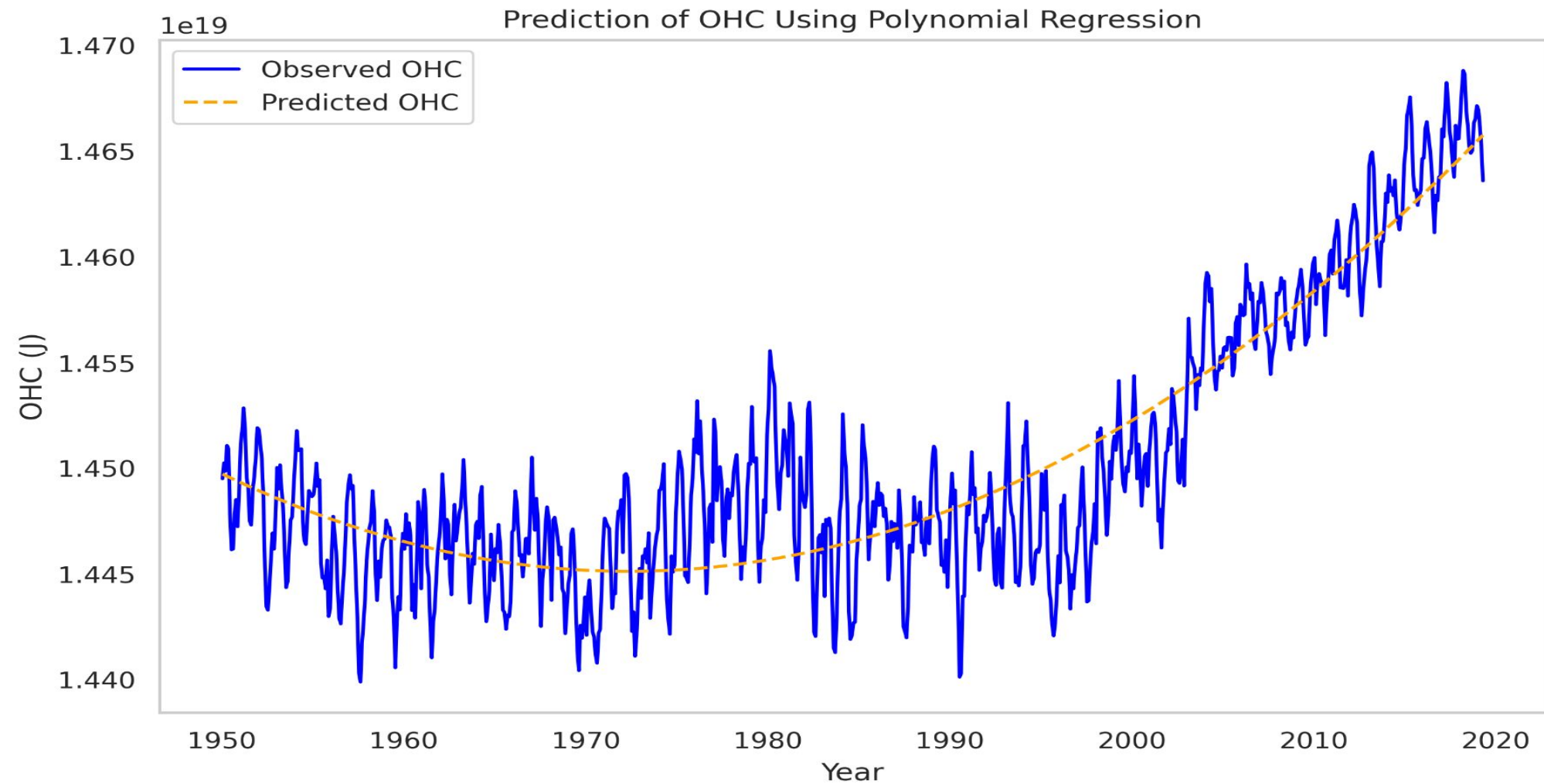


Results: Vertical Structure

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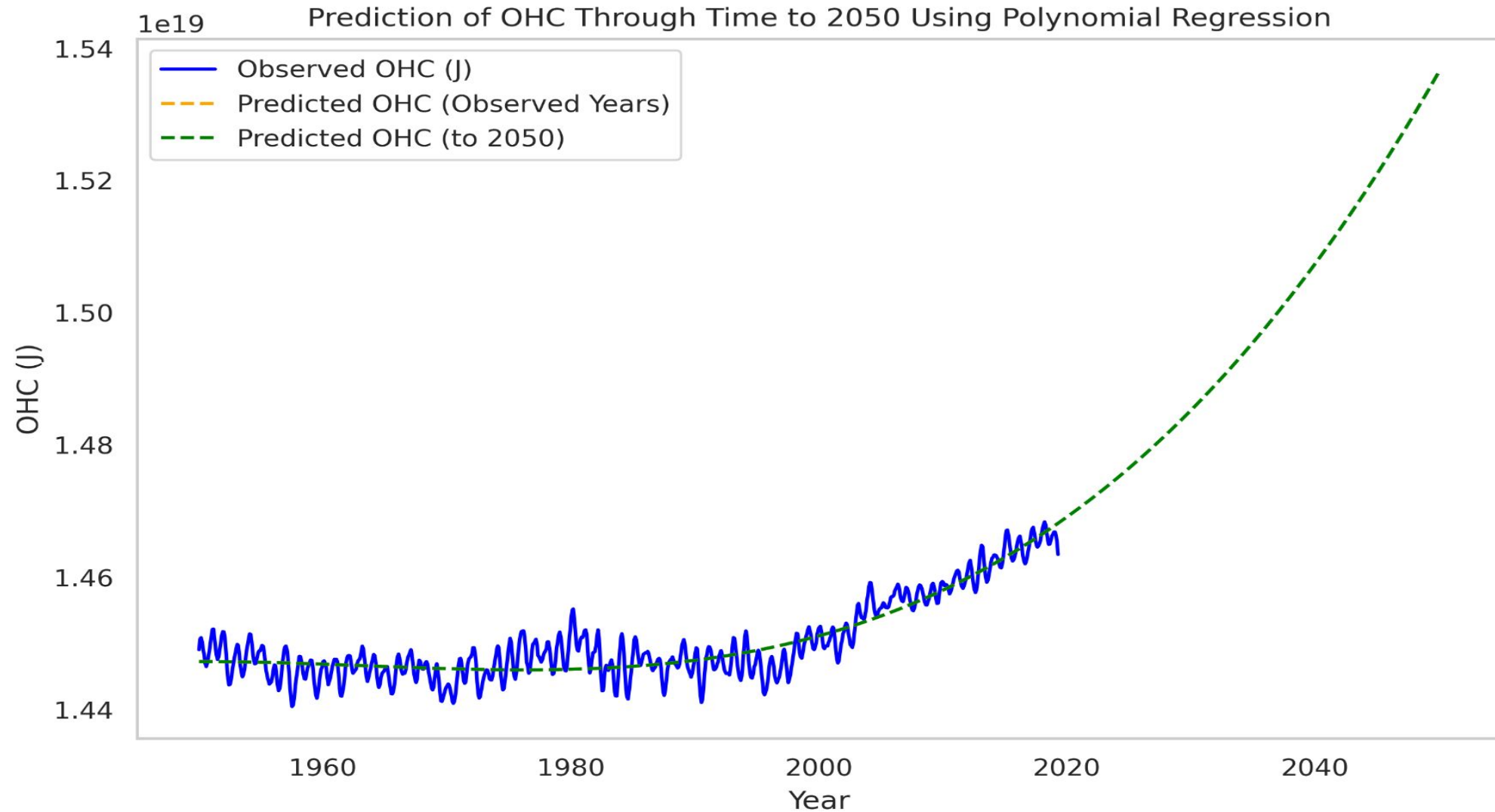


Results: Data Prediction (Polynomial Regression) 8.



Results: Future Prediction

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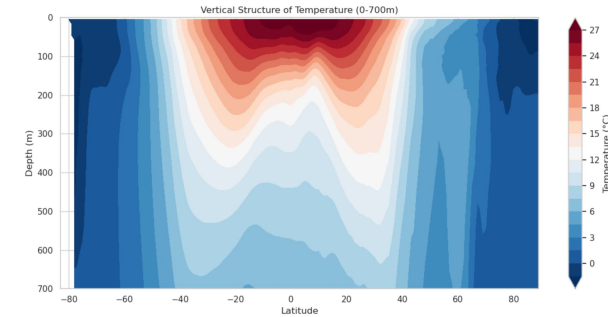


Conclusion:

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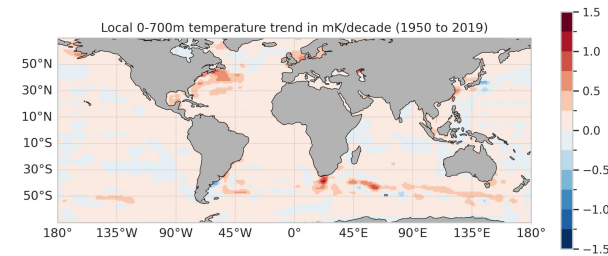
Vertical structure of temperature

- Ocean temperatures exhibit a clear **stratification**, with warmer surface layers and cooler deep layers.
- The **equatorial region** shows the highest surface temperatures, while polar regions experience rapid cooling with depth.



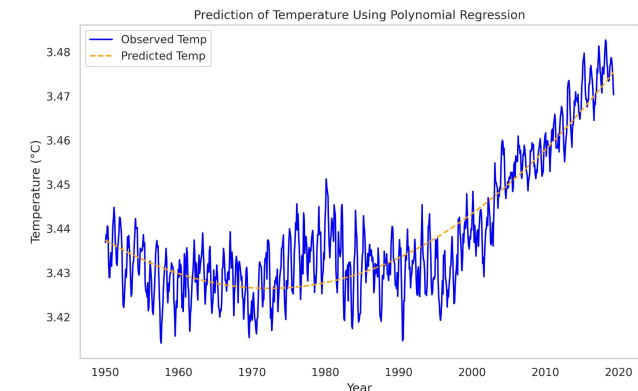
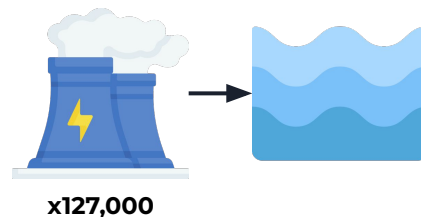
Where global warming affects most the ocean ?

- Hotspots of warming are linked to **western boundary currents** and regions of high **anthropogenic influence**.



Evolution/prediction of OHC and temperature over time

- Energy accumulated by ocean per year : **4×10^{21} Joules**
- **≈ to 127,000 nuclear power plants** (with a production of 1 Gigawatt)

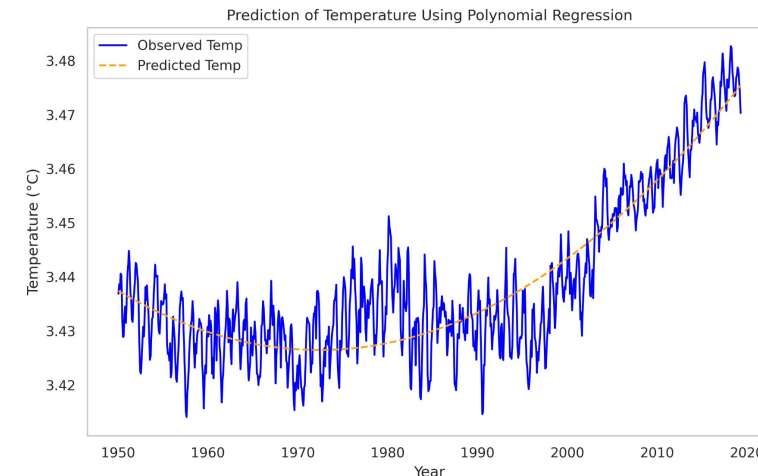
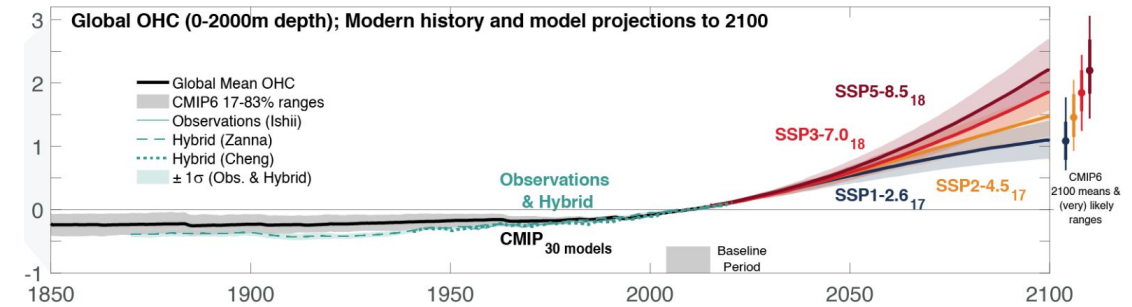
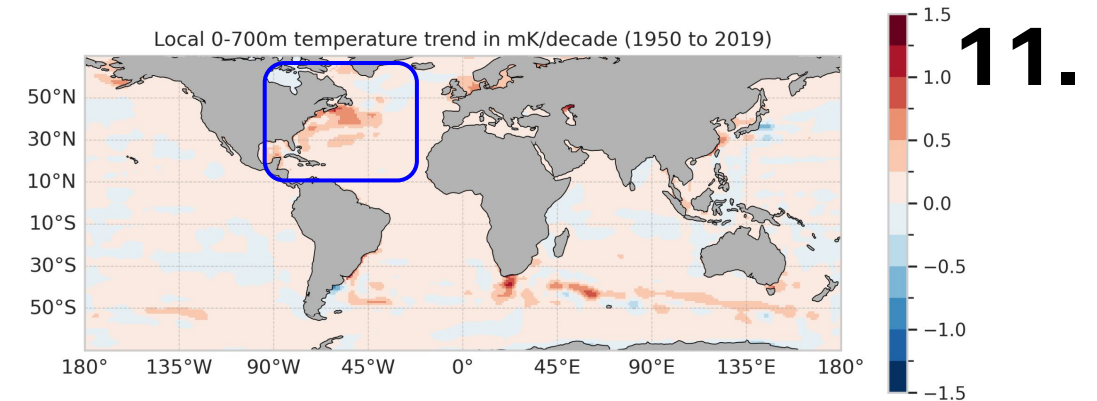


Perspectives

We wanted to focus our study on a specific local area to better understand regional variations in ocean warming.

What scenario do our results align with when compared to the CMIP6 projections?
What about using over models of data ?

Prediction of temperature through time to 2050 using polynomial regression



12.

The warming of the oceans is a stark indicator of the escalating climate crisis



References

13.

Copernicus Marine Service (n.d.). Global Ocean Heat Content (0-2000m) Time Series and Trend. Available at: <https://marine.copernicus.eu/access-data/ocean-monitoring-indicators/global-ocean-heat-content-0-2000m-time-series-and-trend> (Accessed: 22 January 2025).

IPCC (2021a). Chapter 9: Ocean, Cryosphere and Sea Level Change. In: *Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Available at: https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Chapter_09.pdf#page=35 (Accessed: 22 January 2025).

IPCC (2021b). Chapter 9: Ocean, Cryosphere and Sea Level Change (Figure Data). In: *Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Available at: https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Chapter_09.pdf#page=227 (Accessed: 22 January 2025).

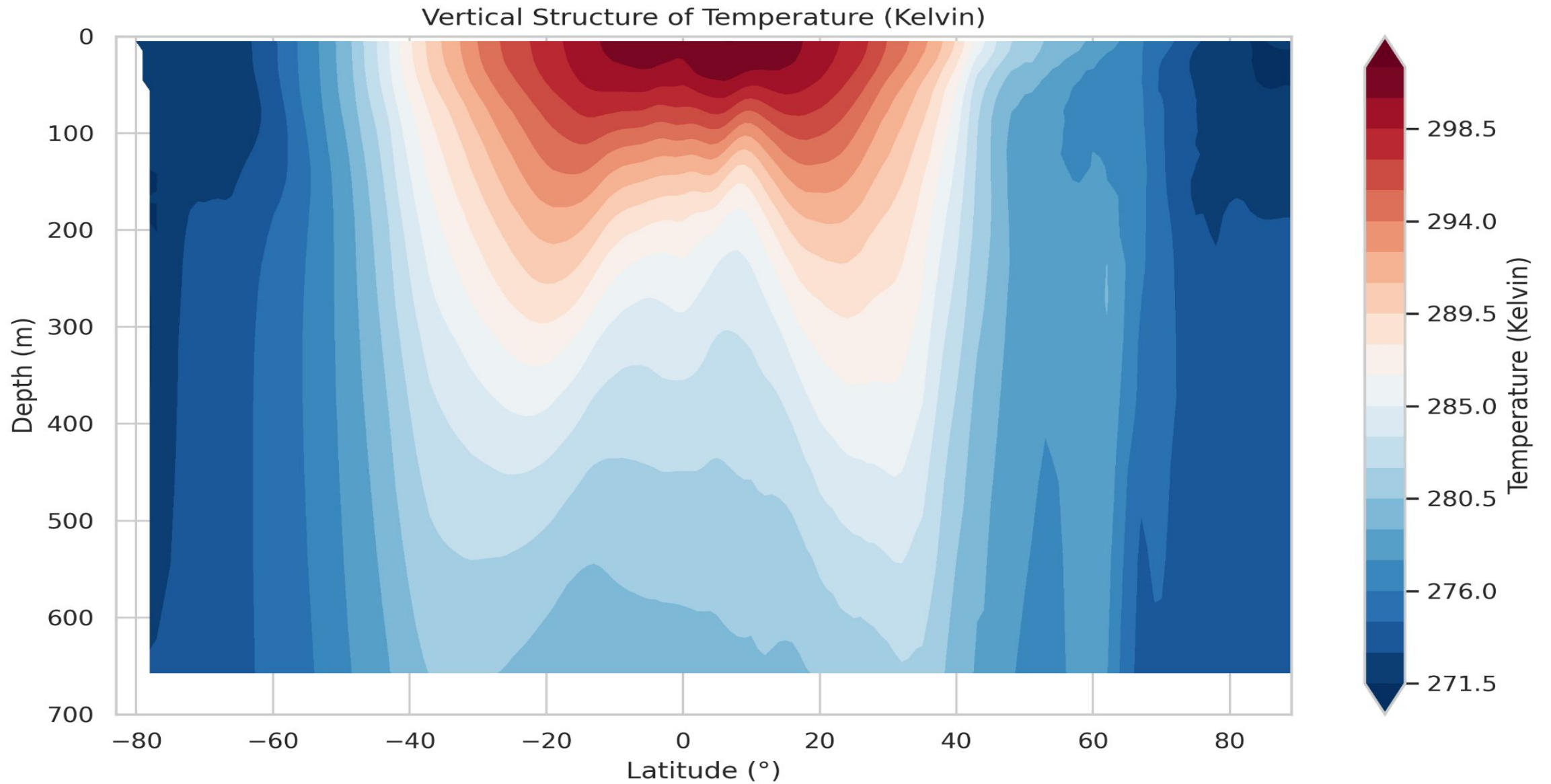
National Centers for Environmental Information (NCEI) (n.d.). Global Ocean Heat Content. Available at: <https://www.ncei.noaa.gov/access/global-ocean-heat-content/> (Accessed: 22 January 2025).

Ocean-Climate.org. (2019) .*The Ocean: A Heat Reservoir*, by Sabrina Speich. Available at: <https://ocean-climate.org/wp-content/uploads/2020/01/1.-The-ocean-a-heat-reservoir-scientific-fact-sheets-2019.pdf> (Accessed: 22 January 2025).

Thank you !

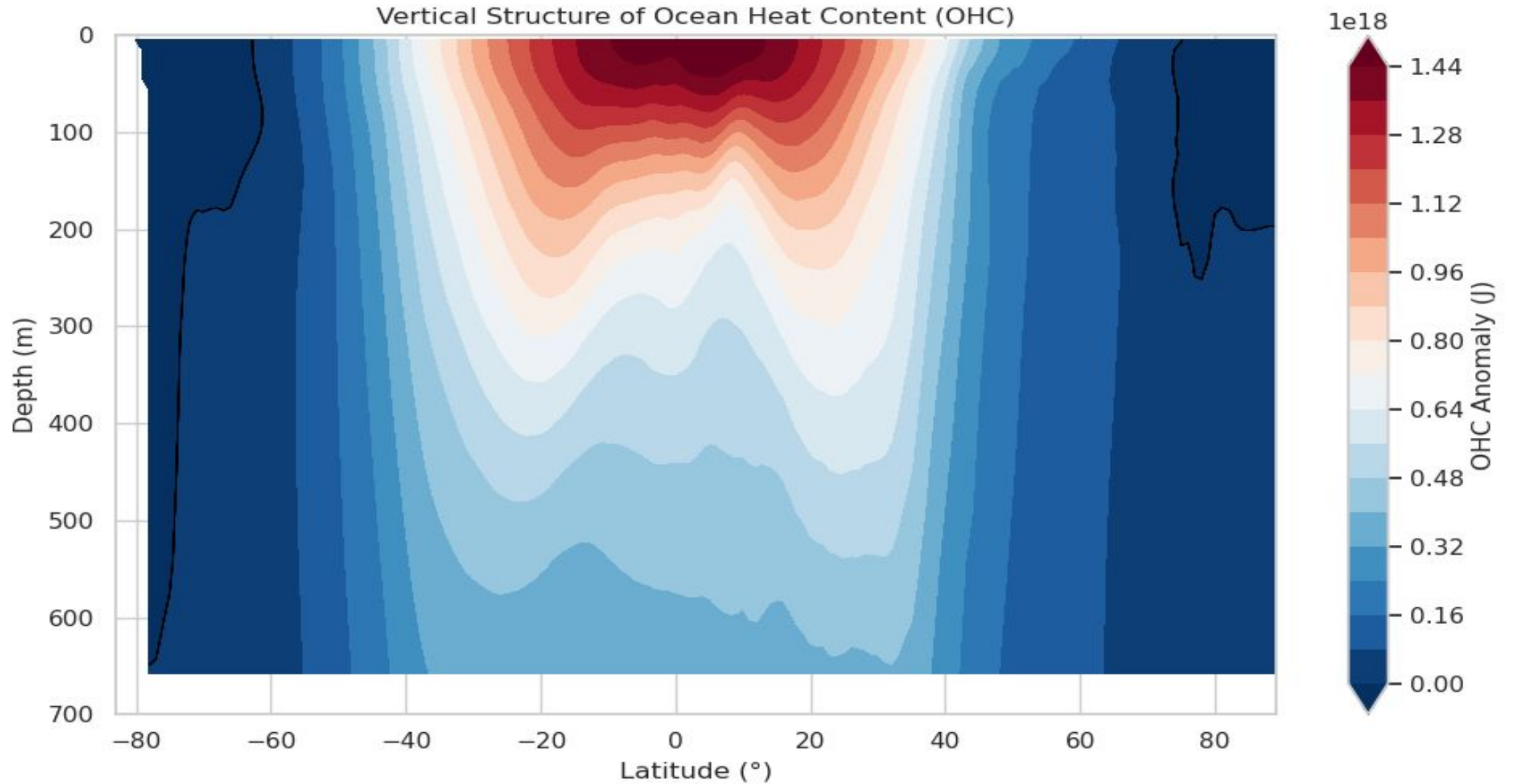
Appendix

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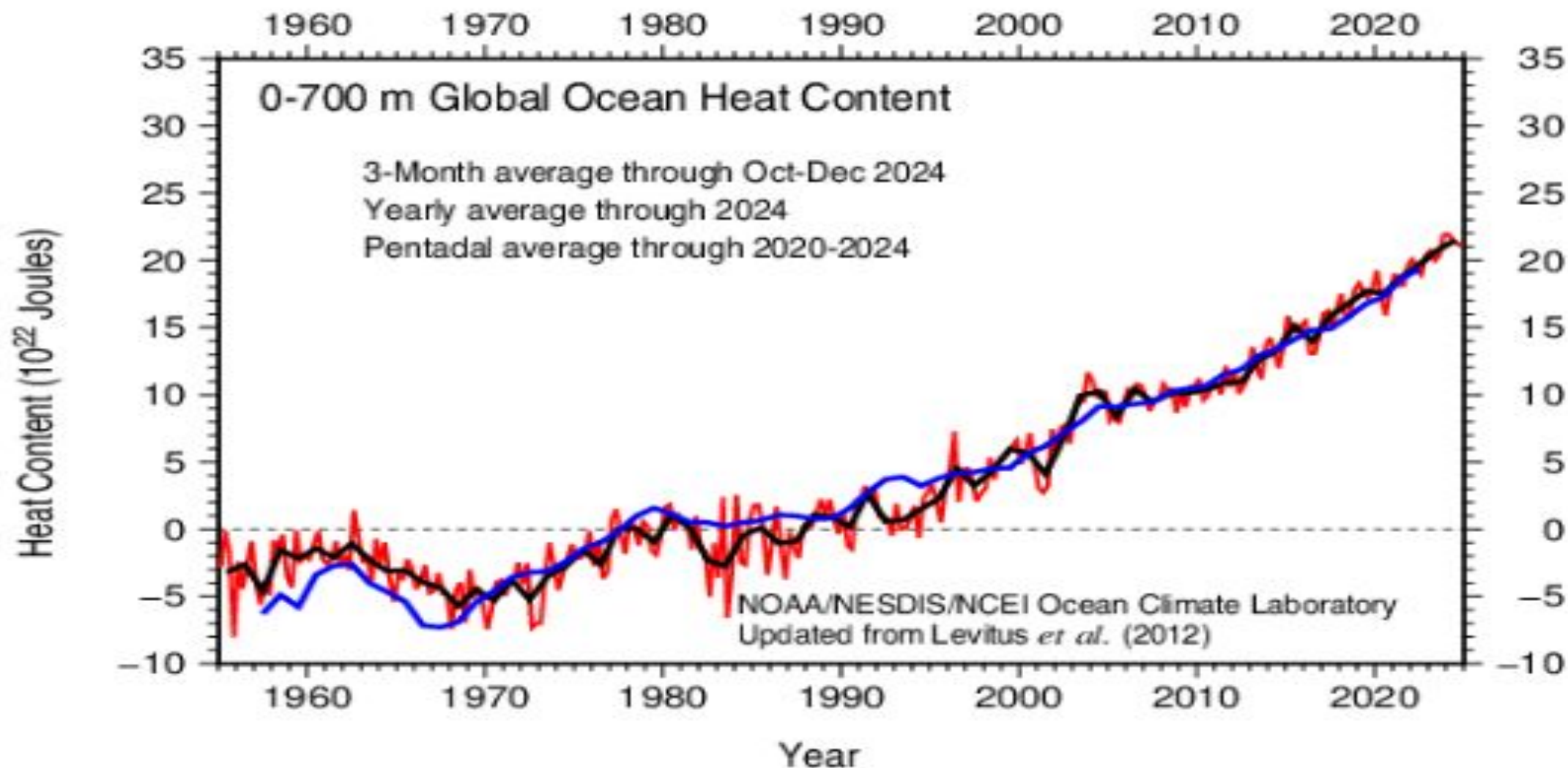
Appendix

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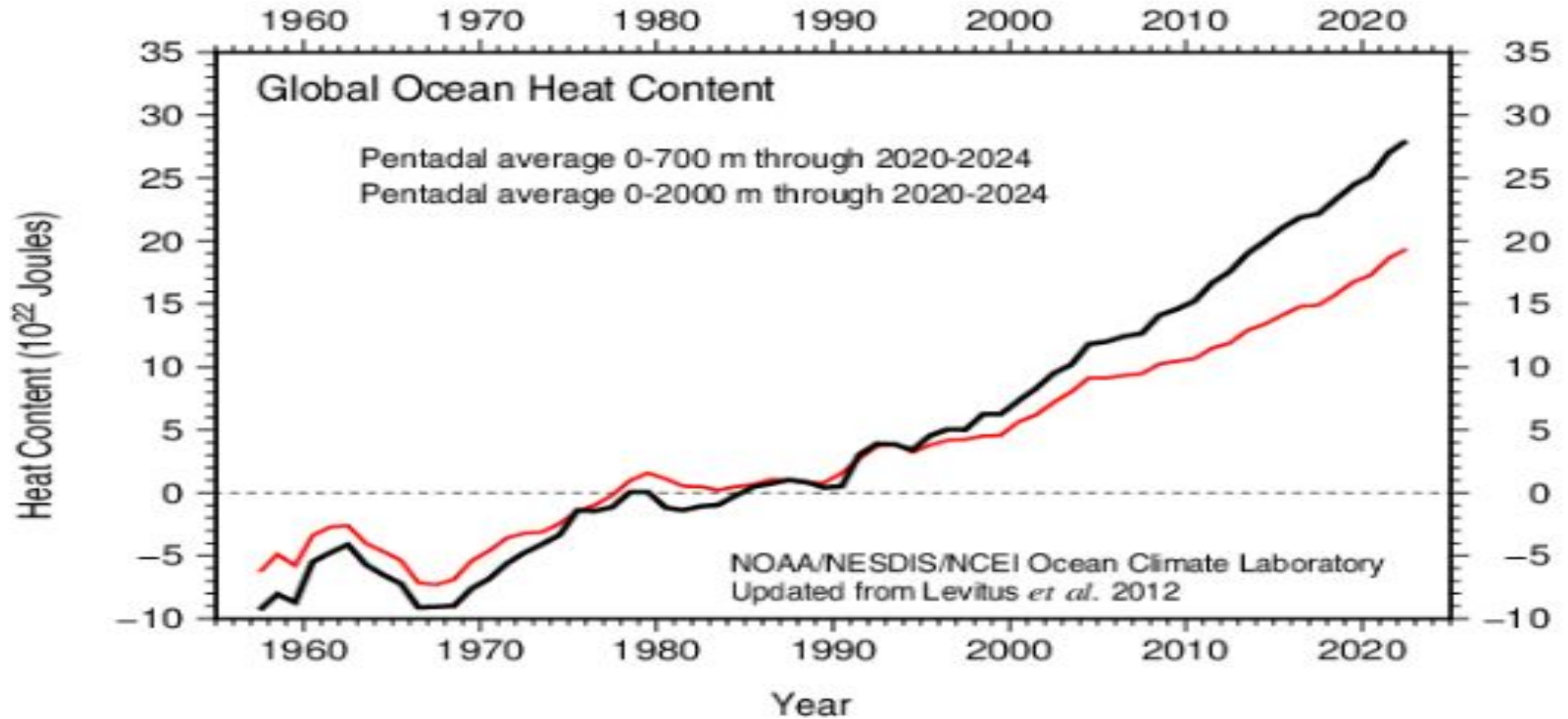


Appendix

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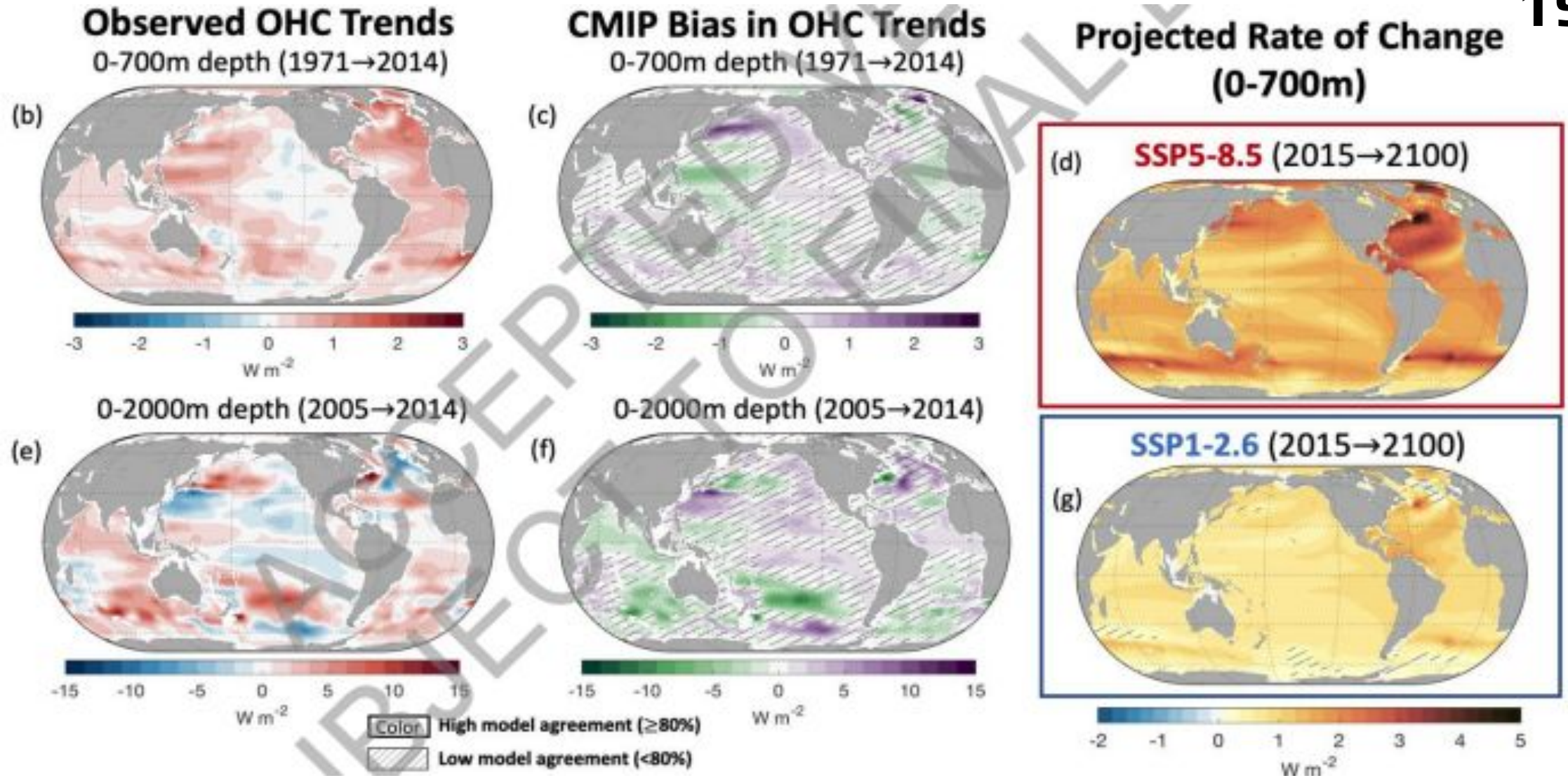
(National Centers for Environmental Information (NOOA), 2025)



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Appendix

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Zonal-mean potential temperature as a function of depth and latitude

Observed mean, CMIP6 biases, and projected rates of change, for each ocean basin

